

Species Classification in High Andean Forests through AIoT Integration

Artificial Intelligence of Things (AIoT) - Juan Manuel Aranda Lopez King

1st Gabriela Reyes Gonzalez
Computer Engineering
Universidad de la Sabana
Chía, Cundinamarca
gabrielarego@unisabana.edu.co

2nd Michael Steven Medina Fernández
Computer Engineering
Universidad de la Sabana
Chía, Cundinamarca
michaelmefe@unisabana.edu.co

I. ABSTRACT

Early detection of invasive species is an important challenge in high Andean forest ecosystems due to the impact these species generate on biodiversity, soil structure, and ecological balance. This work presents an AIoT (Artificial Intelligence of Things) system for the automatic classification of native and invasive plant species using an ESP32-S3 microcontroller with edge processing capabilities.

The system integrates an OV3660 camera to capture images from the environment, an artificial intelligence model trained using Edge Impulse based on a MobileNetV2 architecture, and a cloud monitoring platform using ThingSpeak. The model was trained with a dataset of 1200 images collected in the Neusa Reservoir Park, corresponding to six plant species of the high Andean forest, three invasive and three native.

The obtained results show that it is possible to run image classification models directly on low-power microcontrollers using TensorFlow Lite, achieving an approximate accuracy of 85% in experimental tests. Finally, a web interface was implemented that allows real-time visualization of the detections performed by the device and consultation of the classification history.

This work demonstrates the feasibility of using low-cost AIoT systems for environmental monitoring and the automatic detection of invasive species in natural ecosystems.

II. INTRODUCTION

High Andean forest ecosystems play a fundamental role in water regulation, soil protection, and biodiversity conservation. These ecosystems act as transition zones between mountain forests and páramos, hosting a great diversity of flora and fauna. However, in recent decades, an increase in the presence of invasive species has been observed, significantly altering the ecological dynamics of these environments.

Invasive species can cause the displacement of native species, modifications in soil structure, and changes in the ecological cycles of the ecosystem. In addition, the manual control of these species is complex due to the extent of the territories and the difficulty of constant field monitoring.

In this context, the use of artificial intelligence technologies and the Internet of Things (AIoT) represents a promising alternative for automatic ecosystem monitoring. AIoT systems make it possible to combine sensors, edge computing, and cloud platforms to perform real-time analysis directly on low-power devices.

Initially, the problem was approached through the detection of invasive species in a single image. However, due to the limited resolution of the microcontroller camera (96x96 pixels), it was observed that detection did not allow the characteristics of each species to be correctly distinguished, since much of the vegetation presented similar green tones. For this reason, the problem was reformulated as a classification model of individual species.

The objective of this work is to develop an AIoT system capable of classifying plant species from the high Andean forest using a machine learning model executed directly on an ESP32-S3 microcontroller. The solution integrates image capture, edge inference, cloud storage, and visualization through a web interface.

The main contributions of this work are:

- The construction of an image dataset of native and invasive species of the high Andean forest.
- The training of a classification model optimized for microcontrollers using Edge Impulse.
- The implementation of a complete AIoT system based on ESP32-S3 for real-time inference.
- The development of a monitoring platform based on ThingSpeak and a web interface for visualization of results.

III. PROPOSED SOLUTION

The solution consists of an AIoT system for the automatic classification of plant species in high Andean forest ecosystems. The system integrates components for data acquisition, edge processing, cloud storage, and result visualization.

The general architecture of the system is shown in Figure 1, where the interaction between the different components of the system can be observed. The flow begins with the capture

of images from the natural environment through the camera integrated into the ESP32-S3 Sense device. These images are processed locally using a classification model based on convolutional neural networks (CNN).



Fig. 1. Architecture

Once inference is performed, the device determines whether the detected species corresponds to a native or invasive species. This information is visually represented through two indicator LEDs: a green LED for native species and a red LED for invasive species.

Subsequently, the inference results are sent to the ThingSpeak platform, where they are stored and used as an intermediary between the device and a monitoring web interface. On this platform, three main variables are recorded: the type of detected species (Field 1), the confidence level of the model (Field 2), the identifier of the classified species (Field 3), and a trigger for the activation of image scanning.



Fig. 2. Fields in ThingSpeak

Finally, a web page developed for the project periodically queries the information stored in ThingSpeak and allows visualization of detection history, system status, and ecological recommendations associated with each identified species.

This architecture makes it possible to implement a distributed environmental monitoring system capable of operating with low-cost, low-energy-consumption devices.

IV. IMPLEMENTATION DETAILS

A. Dataset Construction

For training the artificial intelligence model, a dataset composed of 1200 images collected in the Neusa Reservoir Park, located in the municipality of Cogua/Tausa, Cundinamarca, was built. This park presents representative characteristics of the high Andean forest ecosystem and contains both native and invasive species.

During a fieldwork session, approximately 200 photographs were collected for each selected species. The species considered in this study were:

- Invasive species: gorse, patula pine, and bracken fern.
- Native species: borrachero, duraznillo, and siete cueros.



Fig. 3. Species used in the model

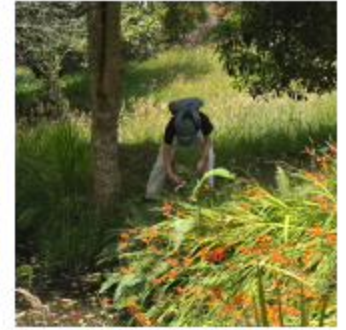


Fig. 4. Dataset collection

The images were captured under different environmental conditions, including variations in lighting, plant size, presence or absence of flowers, and different capture angles. This diversity made it possible to improve the model's generalization capacity during training.

B. Preprocessing

Subsequently, all images were processed using a script developed in Python that allowed them to be automatically resized to a resolution of 96x96 pixels, compatible with the memory restrictions of the ESP32-S3 microcontroller. This process also made it possible to organize and rename the files to facilitate their upload to the Edge Impulse platform.

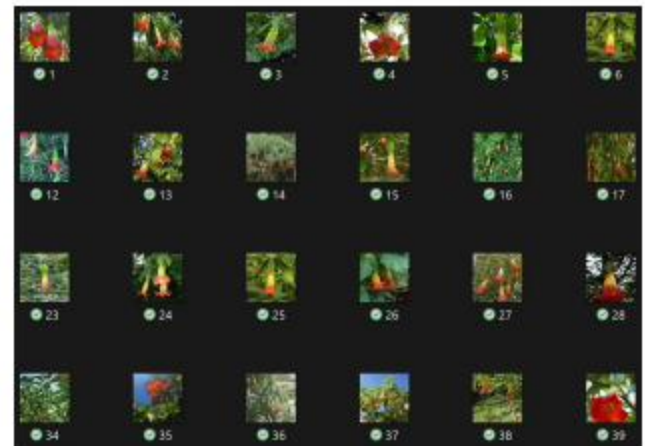


Fig. 5. Resizing of the Borrachero dataset

After preprocessing, the dataset was uploaded to the Edge Impulse platform, where the data was split into training and test sets using a proportion of 80% for training and 20% for validation.

C. Feature Extraction

For the construction of the model, the Edge Impulse image processing block with RGB input was used, since color is a relevant feature for differentiating plant species, especially in the shape and tone of flowers and leaves.

Each input image of 96x96 pixels generates a total of 27,648 features that are subsequently processed by the neural network.

D. Artificial Intelligence Architecture

Initially, a conventional classification model was tested, which achieved an approximate accuracy of 42%. Later, transfer learning was used with a MobileNetV2 architecture of 16 neurons optimized for embedded devices.

The main training parameters were:

- 50 training epochs
- Learning rate of 0.0005
- Data augmentation enabled
- Batch size of 32
- Validation using 20% of the dataset

Each parameter was selected to improve the model, such as the use of 50 epochs to achieve sufficient learning from each image without reaching memorization and avoiding overfitting. Likewise, data augmentation was used to increase the dataset through illumination changes and rotation of preexisting images in the dataset.

The use of transfer learning made it possible to take advantage of previously learned features related to textures, shapes, and visual patterns, which significantly improved the model's performance.

E. Model Deployment

The final model was exported from Edge Impulse as an Arduino library based on TensorFlow Lite, which was integrated into the Arduino IDE environment and executed on the XIAO ESP32-S3 Sense device. For deployment, the quantized version of the model in int8 format was selected, since this optimization significantly reduces memory consumption and improves execution efficiency on embedded devices.

The system was configured to capture images using the OV3660 camera, process them using the classification model, and run inferences directly on the microcontroller. According to the performance estimates provided by Edge Impulse, the quantized model presents an approximate total latency of 1,042 ms and a RAM consumption of 353.1 KB, values considerably lower compared to the non-optimized float32 version, which requires approximately 1.1 MB of RAM and a latency of 1,564 ms. This optimization makes it possible to run the model efficiently on the microcontroller together with the other system components.

During the scanning process, the device performs multiple inferences for approximately ten seconds, finally selecting

the prediction with the highest confidence level as the final classification result.



Fig. 6. Model deployment results

V. EXPERIMENTAL RESULTS AND ANALYSIS

During the model training stage, an approximate accuracy of 70.3% was obtained, with a loss value of 1.89. This result corresponds to the initial performance of the model while it learns to identify the visual characteristics of the different species present in the dataset. The analysis of the confusion matrix at this stage shows that some classes present greater classification difficulties, especially when they share similar visual patterns.



Fig. 7. Results in Train

For example, the 7cuero species presents a low correct classification rate (5.6%), being confused mainly with helecho (36.1%) and duraznillo (25%), which suggests that these species share visual characteristics that the model is still unable to completely differentiate during training. In contrast, other classes show better performance from this stage, such as retamo, which reaches a correct classification rate of 100%, and pino, with approximately 90.9%, indicating that these species have more distinctive visual features for the model.

The global metrics of the model during this phase reflect moderate performance, with an AUC of 0.95, a weighted precision of 0.76, a recall of 0.70, and a weighted F1-score of 0.64. These values indicate that the model still presents some difficulty in correctly generalizing across all classes during the learning process.

Subsequently, in the validation phase using the testing set, the model achieved a global accuracy of 87.5%, which shows a significant improvement in its classification capability. The metrics obtained at this stage show solid model performance, with an AUC of 0.99, a weighted precision of 0.88, a weighted

recall of 0.88, and an F1-score of 0.88, indicating an appropriate balance between precision and the ability to detect the different classes.



Fig. 8. Model results

The analysis of the confusion matrix at this stage shows that most of the species are correctly classified. Species such as pino, retamo, and 7cuero reach classification rates close to 92.5%, while helecho presents approximately 87.5% accuracy. For its part, borrachero reaches a precision of 85% and duraznillo around 75%, the latter being one of the classes with the greatest classification difficulty due to morphological similarities with other species in the dataset.

The F1-score values by class, which combine precision and recall, show a good overall performance of the model, with pino (0.94), retamo (0.93), and helecho (0.91) standing out as the species with the best performance. This indicates that the model is able to adequately distinguish between native and invasive species, which is the main objective of the system.

These results show that the model manages to correctly identify most of the analyzed species, maintaining a good balance between precision and generalization capability.

The final model was successfully deployed on the ESP32-S3 microcontroller using TensorFlow Lite. The inference tests showed that the device is capable of running the model directly at the edge with an approximate inference time of one second, which is suitable for real-time environmental monitoring applications.

The results obtained by the device are sent to the ThingSpeak platform, where three main variables are recorded:

- Type of species (native or invasive)
- Model confidence level
- Identifier of the detected species

The ThingSpeak platform acts as an intermediary between the microcontroller and the web page developed for the project, allowing real-time visualization of classification results, detection history, and recommendations associated with each species.

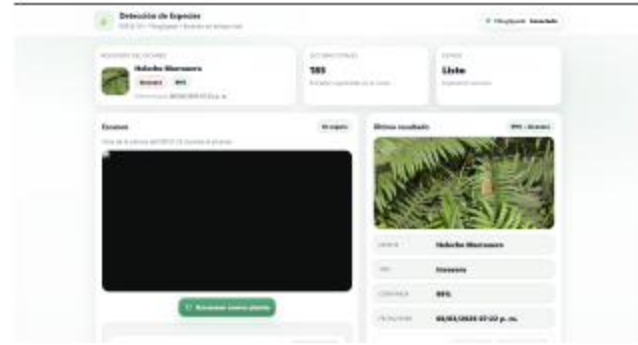


Fig. 9. Developed web page

Overall, the obtained results demonstrate that it is possible to implement artificial-intelligence-based image classification models directly on low-power microcontrollers, integrating edge analysis capabilities (Edge AI) with IoT platforms for data visualization and management.



Fig. 10. Final project test

A. Limitations

Despite the obtained results, the system presents some important limitations that should be considered for future improvements.

First, the resolution of the used images (96x96 pixels) limits the model's ability to capture fine details of the plants. This resolution was selected due to the memory and processing restrictions of the ESP32-S3 microcontroller and the model used for classification. However, by reducing the images to this size, important details such as leaf texture, petal shape, or specific plant patterns are lost, which can generate classification errors when species present similar visual structures.

Second, the size of the used dataset, although sufficient to train a functional model, could be expanded to improve the system's generalization capability. The model was trained with approximately 1200 images, which is a relatively small size for image classification tasks. A larger dataset including greater diversity of environmental conditions, seasons of the year, plant ages, and natural variations of the environment would improve the robustness of the model.

Another important limitation is related to the environmental conditions under which the images are captured. Factors such as abrupt lighting changes, shadows generated by other elements in the environment, plant movement due to wind, or the presence of external objects can affect the quality of the

image captured by the camera and, therefore, the performance of the classification model.

Additionally, some species present morphological similarities in their leaves, flowers, or general structures, which can generate confusion in the model under certain lighting conditions or capture angles. This was evidenced mainly in cases where the plants did not present visible flowers or when the camera captured only partial parts of the plant.

Finally, the system depends on WiFi connectivity to send the results to the ThingSpeak platform. In natural environments or rural areas where connectivity is limited or nonexistent, real-time data transmission may be affected, which would limit the remote monitoring capability of the system.

VI. CONCLUSIONS AND FUTURE WORK

The results obtained in this work demonstrate that AIoT (Artificial Intelligence of Things)-based systems can be effectively used for environmental monitoring in natural ecosystems through low-cost and low-power devices. The implementation of artificial intelligence models directly at the edge makes it possible to reduce dependence on external computing infrastructures and decrease latency in decision-making.

The developed model managed to classify plant species of the high Andean forest using an ESP32-S3 microcontroller with edge processing, which demonstrates the feasibility of running optimized deep learning models on embedded devices with limited resources. The use of transfer learning based on the MobileNetV2 architecture significantly improved the performance of the system compared to a conventional classification model, achieving an accuracy close to 85% in the testing phase.

Likewise, the integration of the system with Internet of Things platforms such as ThingSpeak made it possible to store and visualize the results of the inferences performed by the device. This integration facilitates remote monitoring of the system and enables the construction of visualization interfaces that support environmental analysis and decision-making processes. The developed web page works as a monitoring interface that shows in real time the system status, detection results, and the history of classified species.

Another relevant aspect of this work is the demonstration that artificial intelligence technologies can be used as support tools for ecological monitoring and the detection of invasive species. This type of solution can contribute to environmental conservation processes by facilitating the early identification of species that affect ecosystem balance.

As future work, it is proposed to expand the dataset used for training the model through the collection of a greater number of images under different environmental conditions, seasons of the year, and plant growth stages. This would improve the model's generalization capability and increase its accuracy in real scenarios.

Additionally, it is proposed to incorporate new plant species into the system, both invasive and native, in order to expand the model's scope and improve its usefulness for ecological studies in different ecosystems.

Another future line of work consists of improving the system architecture through the use of more advanced models optimized for embedded devices, as well as exploring additional optimization techniques or more efficient vision models.

It is also proposed to integrate the system with distributed sensor networks that allow multiple devices to be deployed at different points in the ecosystem, thus generating an environmental monitoring network capable of covering larger areas and collecting data continuously.

Finally, the system could evolve into more complete environmental monitoring platforms that integrate multiple sources of information to obtain a deeper understanding of ecosystem status. One possible extension consists of incorporating fauna detection through additional computer vision models capable of identifying birds, mammals, or other organisms present in the ecosystem. This would make it possible not only to analyze the presence of invasive plant species, but also to study their impact on local biodiversity.

Additionally, the system could be complemented with environmental sensors capable of measuring variables such as temperature, relative humidity, solar radiation, or soil moisture. The integration of these sensors with the vision system would make it possible to correlate environmental conditions with the presence or proliferation of certain species, generating valuable information for ecological studies and ecosystem management strategies.

Another line of development consists of integrating the system with geographic information system (GIS) platforms. By recording the location of devices and the detections made, it would be possible to build distribution maps of invasive species within the ecosystem. These maps could be used by researchers and environmental authorities to identify critical invasion zones, monitor changes in vegetation cover, and plan ecological restoration actions.

Taken together, these extensions would allow the system to evolve into an intelligent environmental monitoring network based on AIoT, capable of collecting, processing, and analyzing data continuously in the field, supporting biodiversity conservation processes and sustainable management of natural ecosystems.

REFERENCES

- [1] Universidad Nacional de Colombia, "Conservación de Ecosistemas en Los Soches," Instituto de Estudios Ambientales (IDEA), Universidad Nacional de Colombia. [Online]. Available: https://idea.unal.edu.co/publica/docs/los_soches/Conservacion_Ecosistemas_Los_Soches.pdf
- [2] SEIDOR, "IoT y desarrollo sostenible: cómo la tecnología ayuda al medio ambiente," SEIDOR Blog. [Online]. Available: <https://www.seidor.com/es-es/blog/iot-desarrollo-sostenible-tecnologia-ayuda-medio-ambiente>
- [3] Intel Corporation, "Edge AI: qué es y por qué es importante," Intel. [Online]. Available: <https://www.intel.la/content/www/xl/es/learn/edge-ai.html>
- [4] WWF Ecuador, "¿Por qué las especies invasoras amenazan la biodiversidad?" Feb. 16, 2022. [Online]. Available: <https://www.wwf.org.ec/?374990/por-que-las-especies-invasoras-amenazan-la-biodiversidad>
- [5] Mongabay Latam, "Especies invasoras sin límites: el grave impacto de la flora y fauna exóticas en América Latina," Mongabay, Feb. 2026. [Online]. Available: <https://es.mongabay.com/custom-story/2026/02/especies-invasoras-impacto-flora-fauna-exoticas-america-latina/>

- [6] Telefónica Tech, "IoT ambiental e IA: la fusión para un entorno inteligente," Telefónica Tech Blog. [Online]. Available: <https://telefonicatech.com/blog/iot-ambiental-e-ia-la-fusion-para-un-entorno-inteligente>
- [7] Edge Impulse, "Run Arduino 2.0," Edge Impulse Documentation. [Online]. Available: <https://docs.edgeimpulse.com/hardware/deployments/run-arduino-2-0>
- [8] Edge Impulse, "End-to-end image classification tutorial," Edge Impulse Documentation. [Online]. Available: <https://docs.edgeimpulse.com/tutorials/end-to-end/image-classification>
- [9] Seeed Studio, "Edge Impulse Vision AI," Seeed Studio Wiki. [Online]. Available: <https://wiki.seeedstudio.com/es/edge-impulse-vision-ai/>